

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 5 · Correlation and basic regression

TUESDAY SEP 22 AND THURSDAY SEP 24, 2026

WHAT YOU WILL BE ABLE TO DO BY MONDAY NIGHT

- 1** Fit a line to two columns of data. Say in your own words what the slope, intercept, and residual mean.
- 2** Read a regression output table and explain what standard error, the confidence interval, and the p value tell you.
- 3** Understand why coefficients can clear 0.05 and still not be worth acting on due to chance, lurking variables, and small effect sizes.



UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials.



3 Confirm that weeks/week05/ now exists in your repo.

It holds this week's README, the worksheet you fill in on Thursday, and the two Case Study data files.

01

CORRELATION AND THE REGRESSION LINE



A TEAM IS 50 AND 31 AFTER 81 GAMES. HOW MANY OF ITS LAST 81 DO YOU EXPECT IT TO WIN?

Say a number, and say what you based it on.

PREDICTION

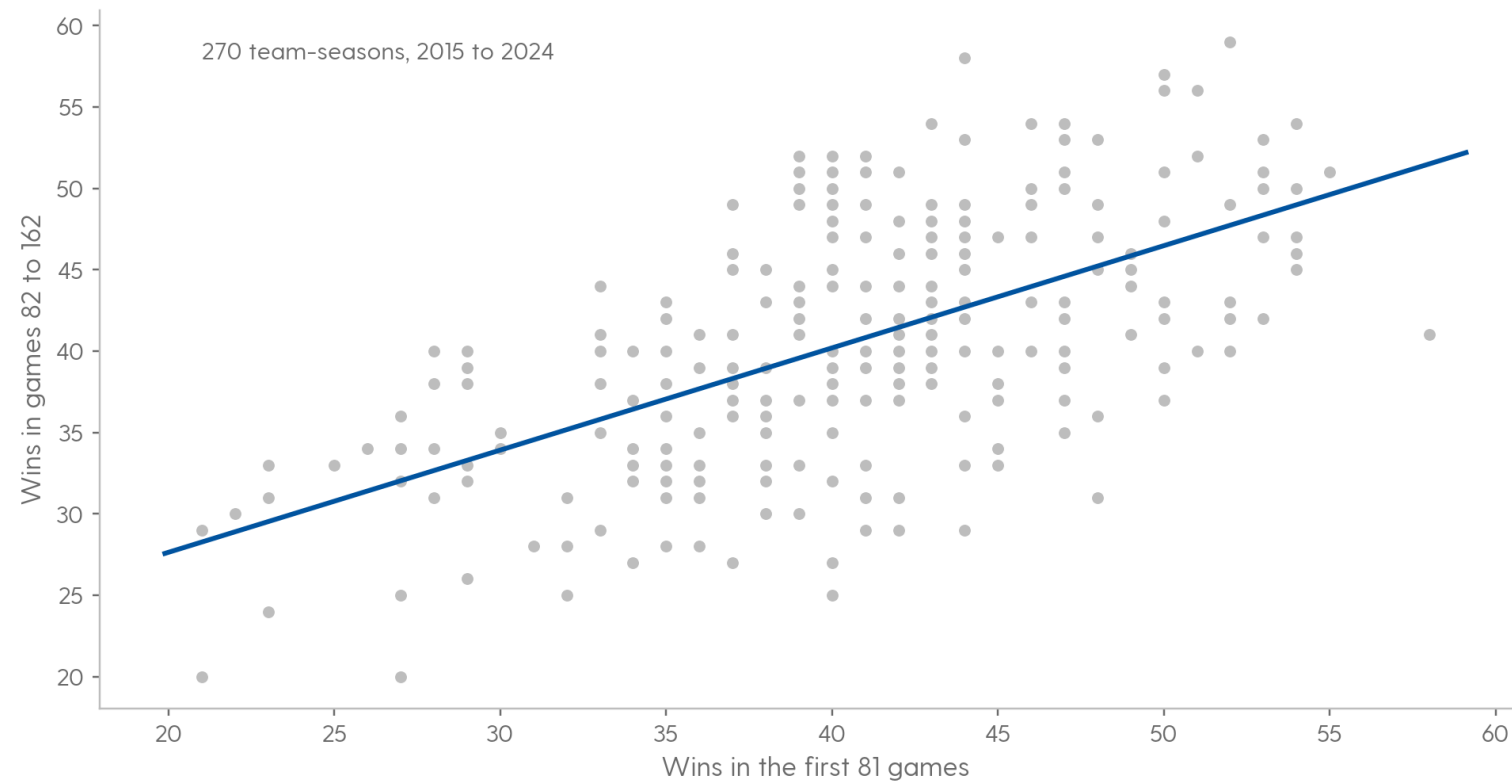
2024 TEAMS, BY HALF

TEAM	FIRST 81 GAMES	WINS IN GAMES 82 TO 162
St. Louis Cardinals	42-39	41
San Diego Padres	41-40	52
Tampa Bay Rays	40-41	40
Houston Astros	40-41	48

All four split the first half within 2 games of each other.

CORRELATION

FIRST-HALF WINS AND SECOND-HALF WINS, 2015 TO 2024



270 team-seasons. The correlation between the two halves is $r = 0.58$.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

CORRELATION (R)

How tightly two columns move together, from -1 to 1. A correlation of 0.58 says the dots cluster around a rising line. It does not say how steep the line is.

$r = 1$ is a perfect straight line upward, $r = -1$ a perfect line downward, $r = 0$ no linear relationship at all. In Excel: CORREL(range1, range2).

THE REGRESSION LINE

$$y = a + b \times x$$

Our prediction of y is a y-intercept (a), plus a coefficient (b) multiplied by our independent variable (x).

prediction		intercept	coefficient		
second-half wins	=	15.1	+	0.63	× first-half wins

In Excel: SLOPE(y range, x range) gives b and INTERCEPT(y range, x range) gives a.

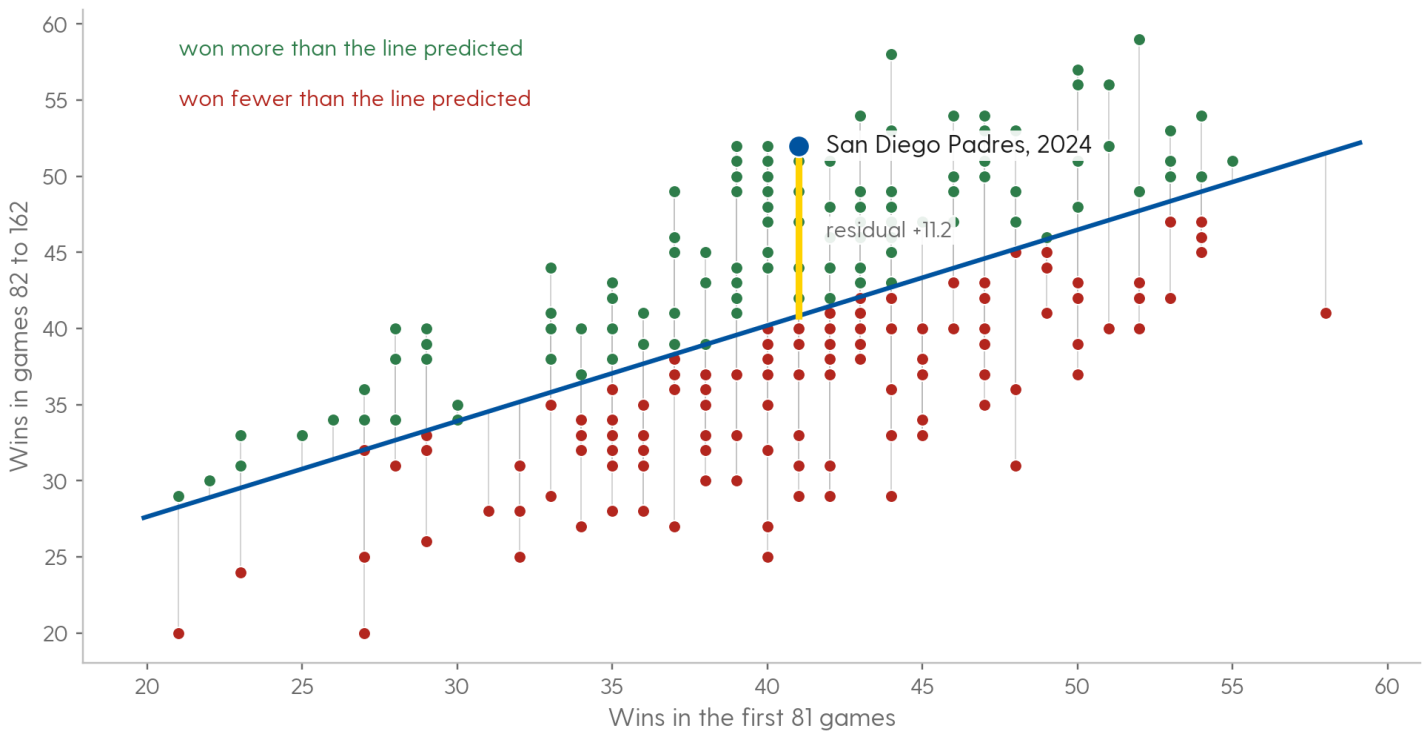


WHAT DOES THE 0.63 MEAN, IN WORDS?

For each increase in ____ (our independent variable), we expect an increase of ____ in ____ (our dependent variable).

RESIDUALS

SAN DIEGO PADRES, 2024



41 wins in the first half, so the line predicted 40.8 in the second. They won 52. The gap of +11.2 is the residual.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

RESIDUAL (ε)

Actual Y doesn't always equal $a + bx$. What's left is "random error" (ε). Sometimes this is due to randomness, other times it is due to factors outside of our model.

residual = actual y minus predicted y . The line is the one that makes the sum of the squared residuals as small as possible, which is why it is called least squares.

R SQUARED

34%

of the variation in second-half wins is accounted for by first-half wins. This is R squared, the correlation squared.

The other 66% is the residuals: injuries, trades, schedule, and luck. In Excel: `RSQ(y range, x range)`.



**WHAT ADDITIONAL FIRST-HALF TEAM DATA POINT COULD BETTER
PREDICT SECOND HALF TEAM PERFORMANCE?**

? IF WE PUT BOTH IN THE SAME MODEL, WHAT R SQUARED DO YOU EXPECT?

PREDICTORS DO NOT STACK.

Two columns that each explain a third of the variation do not explain two thirds together. They explain the same third. Adding a predictor is only worth it when it knows something the others do not.

02

READING THE REGRESSION OUTPUT

THE OUTPUT TABLE

SECOND-HALF WINS ON FIRST-HALF WINS, 270 TEAM-SEASONS

	INTERCEPT	FIRST-HALF WINS	WHAT IT MEANS
Coefficient	15.08	0.63	The slope: each first-half win is worth 0.63 second-half wins
Standard error	2.20	0.054	How much the coefficient would move in a new sample
Lower and upper 95%	10.75 to 19.41	0.52 to 0.73	The range the true coefficient falls in, with 95% confidence
t Stat	6.9	11.7	Standard errors the coefficient sits from zero
P-value	less than 0.0001	less than 0.0001	The chance of a slope at least this far from zero if there were no relationship between first- and second-half wins

This is the table Excel prints from Data Analysis, Regression, or from LINEST(y range, x range, TRUE, TRUE), with its columns down the side. MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

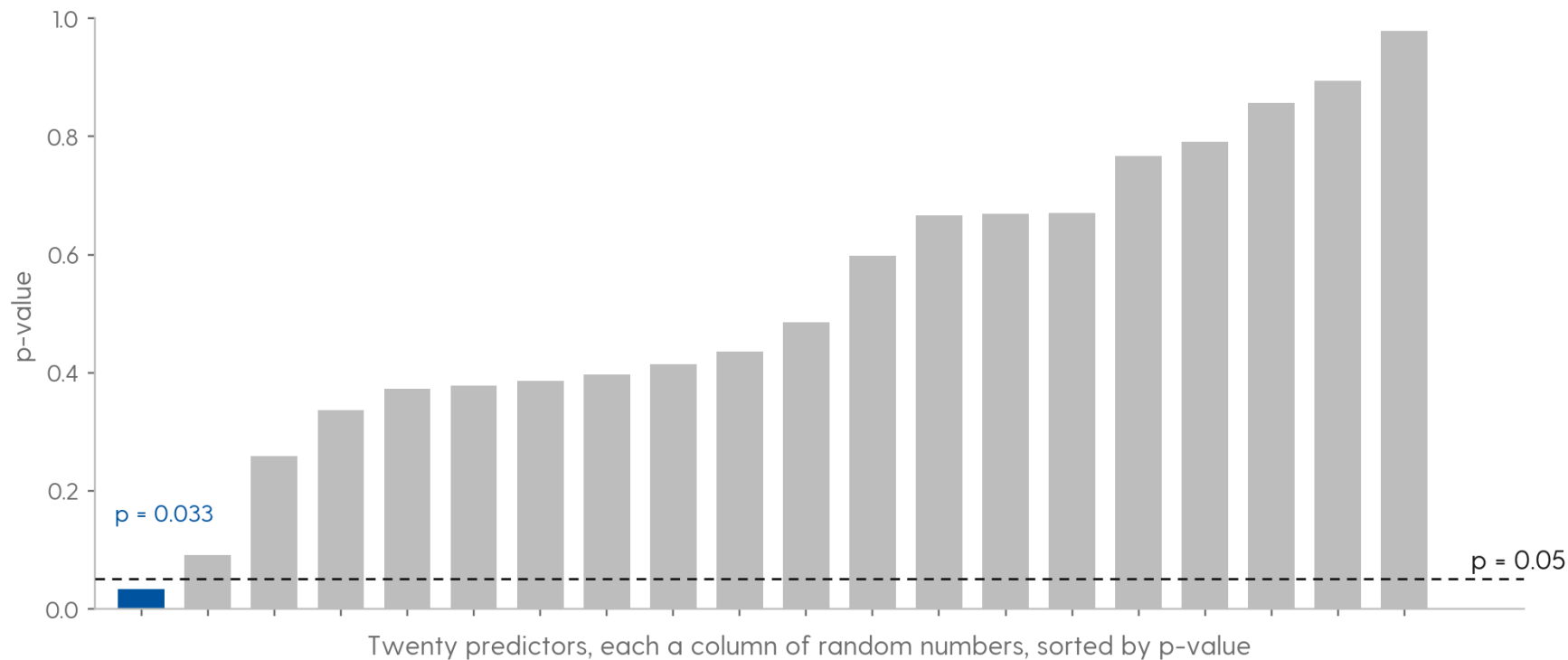
**THE NULL HYPOTHESIS: FIRST-HALF WINS TELL YOU
NOTHING ABOUT SECOND-HALF WINS.**

STATISTICALLY SIGNIFICANT

Typically, 0.05 is the P-Value used for "Statistical Significance". It means: if there were no relationship, a result this far from zero would show up less than one time in twenty.

If we tested twenty completely unrelated variables for correlation with Team Win % (Mascot Height, Letters in Team Name, etc.), we'd expect one of them to return as "statistically significant" by chance.

TWENTY RANDOM PREDICTORS OF 2024 WINS



Twenty columns of random numbers, each regressed on the 30 teams' win totals. 1 of the twenty crossed 0.05. Across 1,000 batches of twenty, at least one crossed in 64% of them.

Random predictors drawn with a fixed seed; wins from the MLB Stats API, 2024 regular season.

**STATISTICS DO NOT REPLACE CRITICAL THINKING OR
UNDERSTANDING THE SPORT!**

P-HACKING

Testing many things and reporting the one that cleared 0.05 as if it had been the only test. The chart before this one is what twenty tests look like when nothing is there.

Every extra predictor you try is another draw. Decide what you are testing before you look, and report how many things you tried.

BEYOND STATISTICAL SIGNIFICANCE

Spurious correlation

Mozzarella eaten per person and engineering bachelor's degrees awarded, $r = 0.97$. Ten years of national data, 2012 to 2021. Two series that both drift upward, and nothing behind it.

Lurking variable

Stork pairs and human birth rates across 17 European countries, $r = 0.62$, $p = 0.008$. It clears 0.05. Bigger countries hold more storks and more people, so country size moves both columns.

03

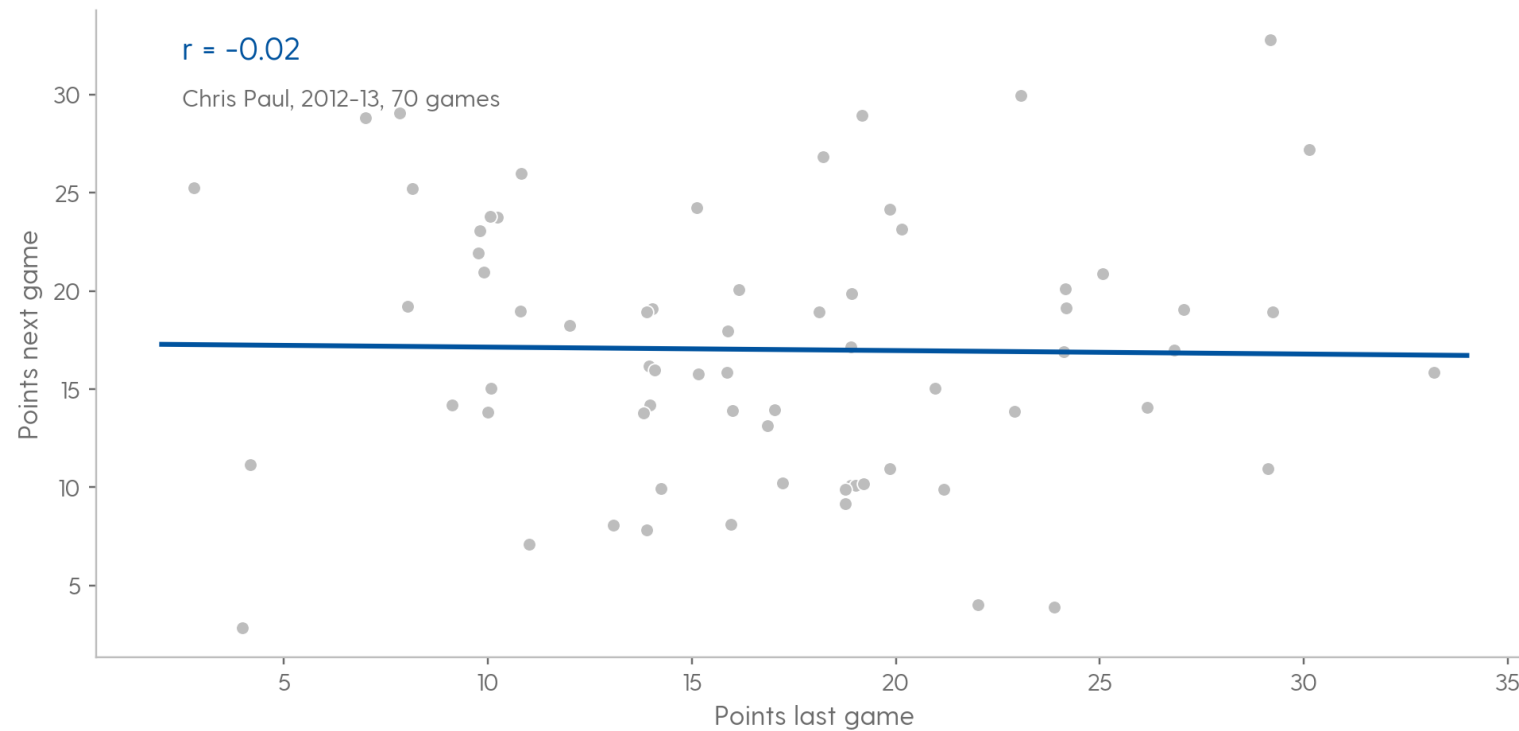
STATISTICAL AND PRACTICAL SIGNIFICANCE



A PLAYER SCORED MORE THAN HIS AVERAGE LAST GAME. SHOULD I EXPECT HIM TO SCORE MORE IN OUR NEXT GAME?

AUTOCORRELATION

CHRIS PAUL, 2012-13



Each dot is one game's points against the next game's. 70 games, $r = -0.018$, $p = 0.89$. A big night tells you nothing about the next one.

Basketball-Reference. Game log, Los Angeles Clippers 2012-13.

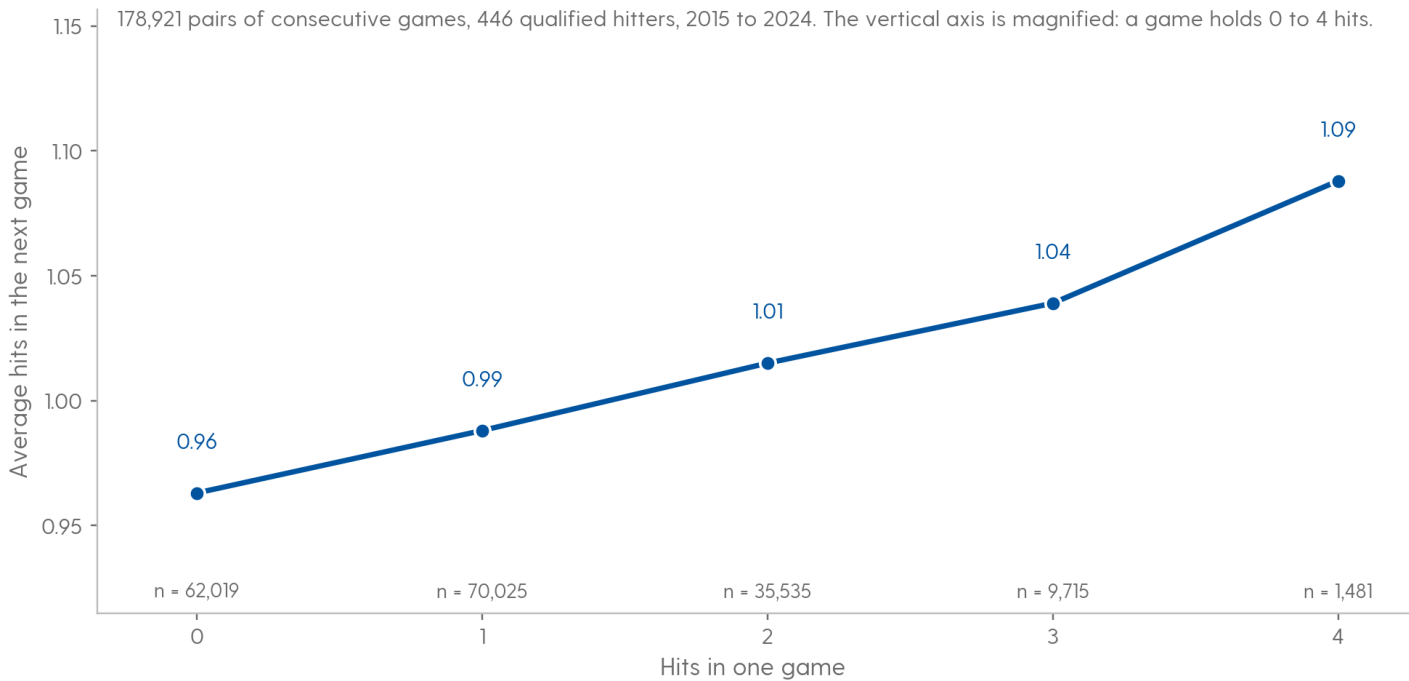
AUTOCORRELATION

The correlation of a series with itself one step later: this game against the next, this month against the next. A streaky player has a positive one.

In Excel: CORREL over two ranges offset by one row. A positive one says hot streaks carry over, a negative one says they correct.

WHO IS BATTING?

HITS IN ONE GAME AND HITS IN THE NEXT, 178,921 PAIRS



The average next game climbs 0.13 hits from a hitless game to a four-hit game. Across all 178,921 pairs the correlation is 0.027, the slope is 0.027 hits per hit, and the p-value is less than 0.0001.

MLB Stats API, game logs for every qualified hitter, regular seasons 2015 to 2024, 2020 excluded.



DOES THAT CONVINCE YOU THAT A HITTER WHO HITS WELL IN ONE GAME WILL HIT WELL IN THE NEXT?

The p-value is as small as any we have seen this week. Say what you would want to check before you believed it.

TWO WAYS OF MEASURING THE SAME PAIRS

	PAIRS	R	P-VALUE	R SQUARED
Every hitter pooled together	178,921	0.027	less than 0.0001	0.071%
Each hitter against his own season average	178,921	0.005	0.044	0.002%

Season averages run from 0.55 hits a game to 1.41 across the 1,233 player-seasons. Pooling lets that gap stand in for a hot streak: good hitters get hits in both games. Both rows clear 0.05, and neither effect is worth acting on.

STATISTICAL VS PRACTICAL SIGNIFICANCE

With enough data, an effect that is nearly zero will still clear 0.05. Significance says the effect is probably not zero. It does not say the effect is big enough to act on, and it does not say you measured the right thing.

Both rows above are significant and neither is useful. The honest version, the bottom row, says a hitter who got three hits yesterday is expected to get 0.010 extra hits today, about one hundredth of a hit. Nothing a manager can put in a lineup. Read the coefficient and R squared first, then ask what else moves both columns.

04

CORRELATION AND CAUSATION



NAME A STATISTIC THAT GOES WITH WINNING WITHOUT CAUSING IT.

CORRELATION $\neq$ CAUSATION

QB Kneels and Wins

Attendance and Win %

Draft Position and Record

Good Vibes and Wins

These aren't happening by random chance, they are meaningfully related. They just aren't directly causal.

LURKING VARIABLE

A third thing that moves both of the two you measured.
Take it out and the relationship you saw can shrink or disappear.

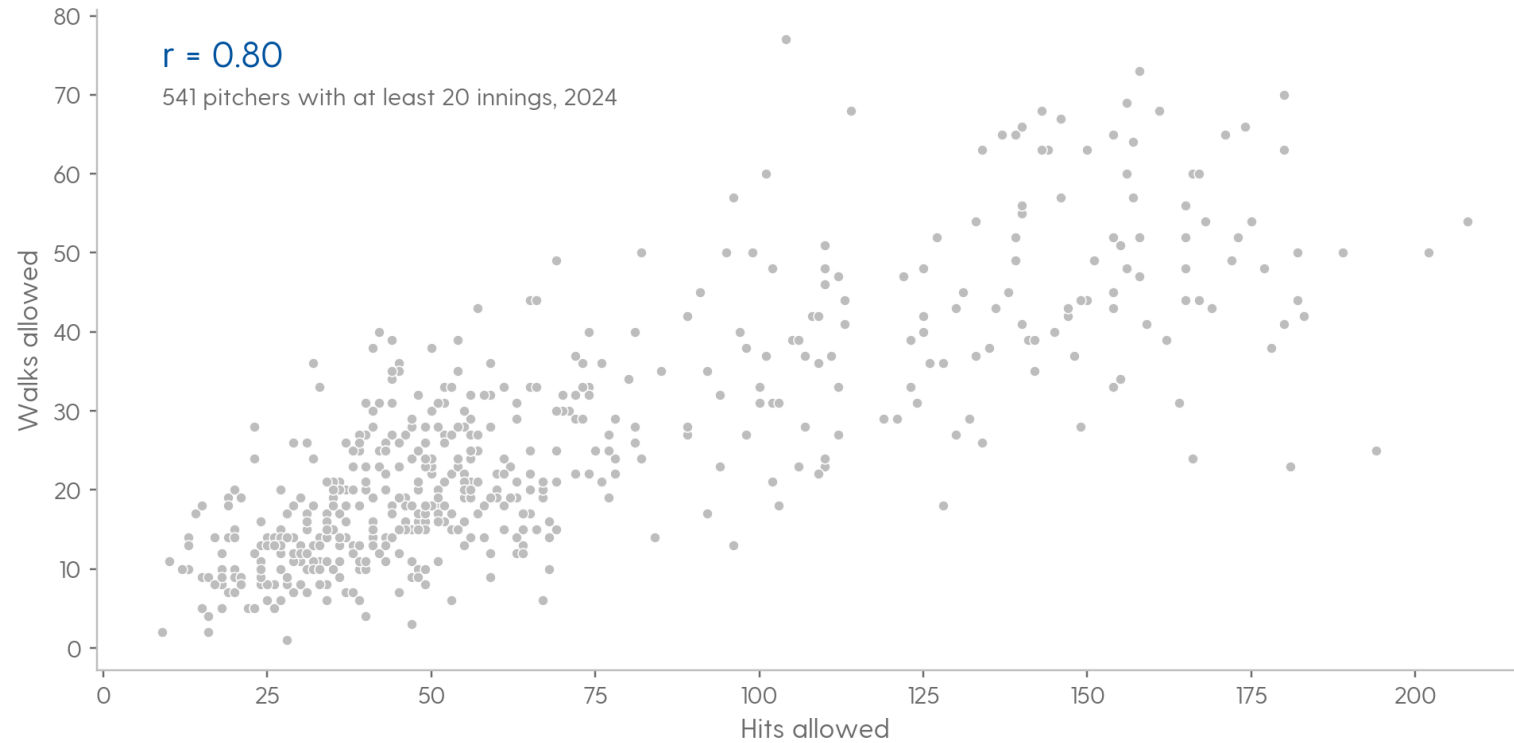
Two ways to take it out: express both variables per unit of the third, or put the third variable in the regression as another predictor.



ARE WALKS ALLOWED AND HITS ALLOWED CORRELATED ACROSS MLB PITCHERS?

Say yes or no, and say how strong you think it would be.

HITS AND WALKS ALLOWED, 541 PITCHERS WITH AT LEAST 20 INNINGS



The two counts correlate at $r = 0.80$. Pitchers who give up hits also walk people.

MLB Stats API, 2024 regular season.

PARTIAL CORRELATION: HITS AND WALKS WITH INNINGS TAKEN OUT

$$r(x, y \mid z) = (r(x, y) - r(x, z) \times r(y, z)) \div \sqrt{((1 - r(x, z)^2)(1 - r(y, z)^2))}$$

x and y are the two things you measured. z is the third thing you suspect moves both of them.

$$r(H, W \mid I) = (r(H, W) - r(H, I) \times r(W, I)) \div \sqrt{((1 - r(H, I)^2)(1 - r(W, I)^2))}$$

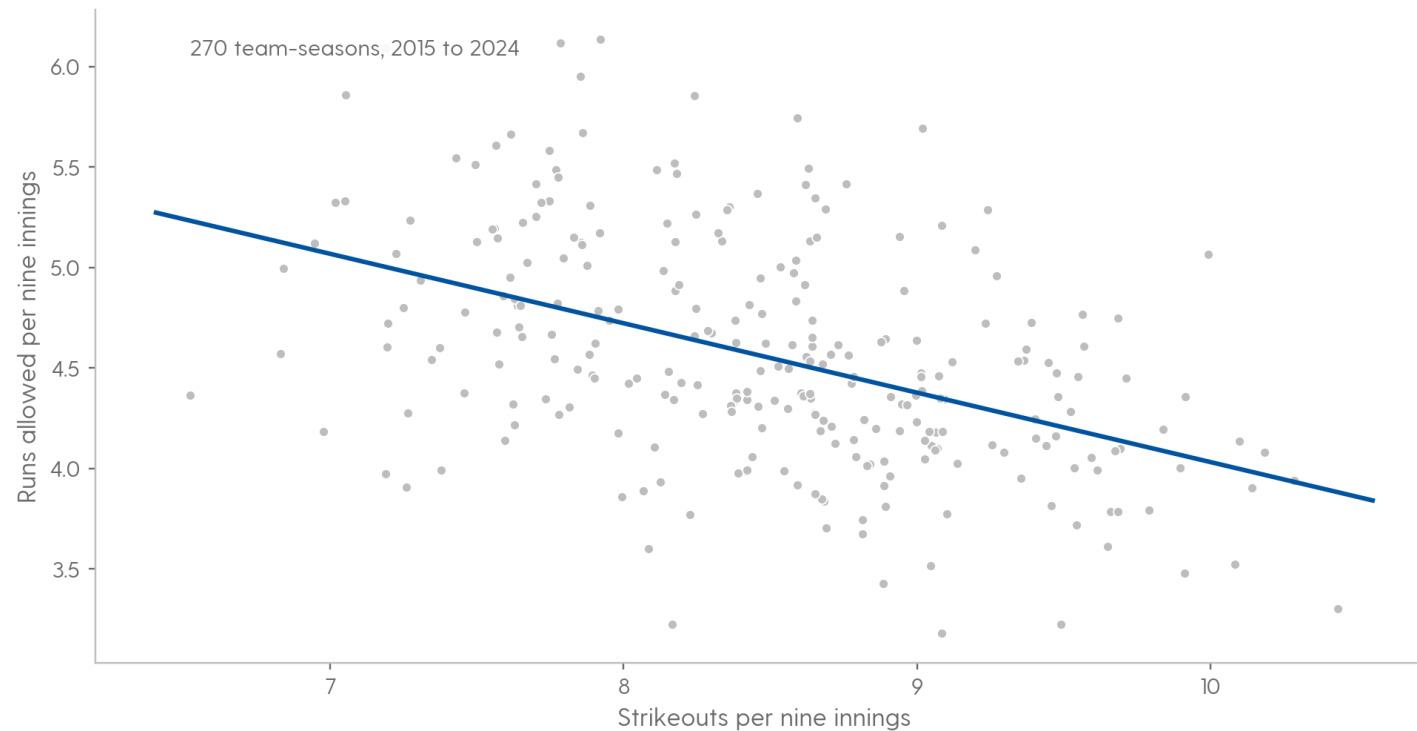
$$= (0.80 - 0.96 \times 0.83) \div \sqrt{((1 - 0.96^2)(1 - 0.83^2))}$$

The correlation between hits and walks once innings pitched is held fixed. Same answer as the per-nine chart, by a different route.

$$R(H, W \mid I) = 0.02$$

? DO STRIKEOUT PITCHERS ALLOW FEWER RUNS? WHAT ELSE WOULD YOU WANT IN THAT MODEL?

TEAM RUNS ALLOWED AND STRIKEOUTS PER NINE, 270 TEAM-SEASONS

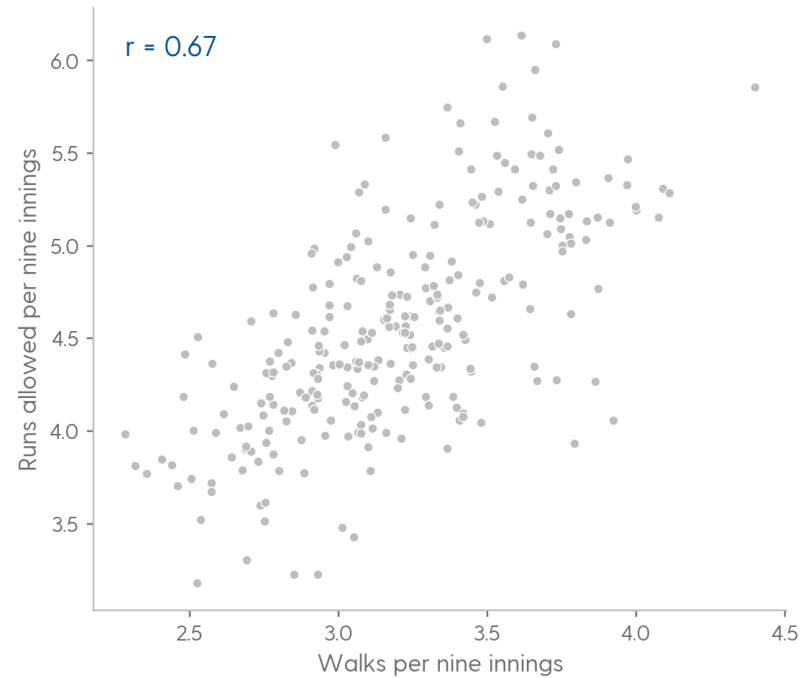
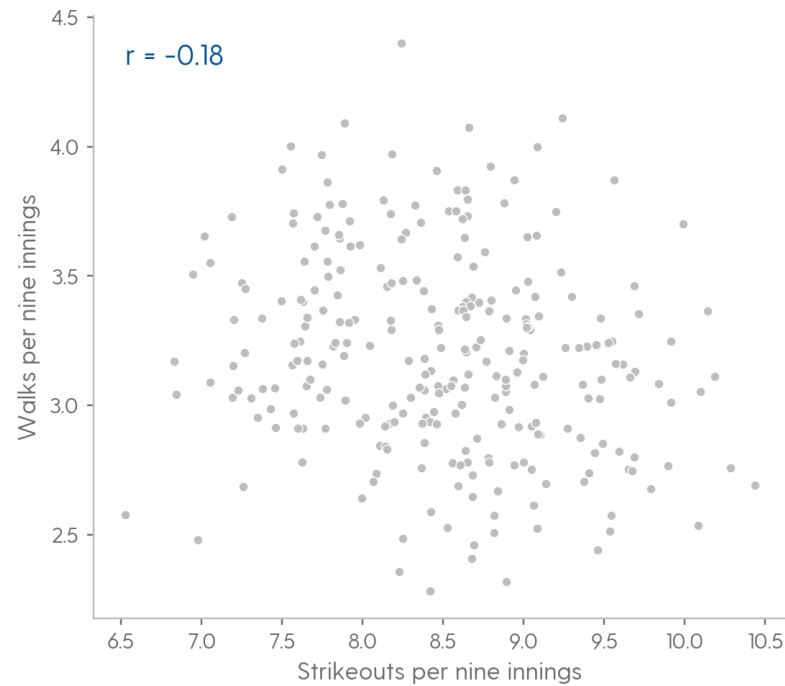


They correlate at $r = -0.46$, and the line says each strikeout per nine is worth -0.35 runs. On its own the model explains 21% of the variation.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

LURKING VARIABLES

WALKS PER NINE, AGAINST STRIKEOUTS AND AGAINST RUNS



270 team-seasons, 2015 to 2024

Walks barely track strikeouts at $r = -0.18$, and they track runs allowed hard at $r = 0.67$. A team's walk rate is most of what the strikeout model was missing.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

05

RELATIVE IMPORTANCE



TEAM POINTS PER GAME ON EFFECTIVE SHOOTING PERCENTAGE AND SHOT ATTEMPTS. WHICH COEFFICIENT WILL BE BIGGER, AND WILL THAT MEAN IT MATTERS MORE?

NBA TEAM POINTS PER GAME, 2019-20 TO 2024-25

$$\text{points per game} = -73.2 + 182 \times \text{eFG\%} + 0.99 \times \text{FGA}$$

eFG% runs from 0 to 1, so the 182 is per whole unit: 1.8 points per percentage point.

FGA is attempts per game, so the 0.99 is 0.99 points per extra shot.

180 team-seasons, R squared 0.89. eFG% counts a three as one and a half makes, so it credits shot quality as well as accuracy. Both coefficients clear 0.05 easily, and the two numbers cannot be compared to each other: one is per unit of a percentage, the other per shot.

06

THIS WEEK'S WORK



WEEK 8 OF THE 2021 NFL SEASON

- 1 Open hand-build-regression-worksheet.xlsx and its first sheet, slope by hand.
- 2 Build the slope the long way from the deviations, then check it with SLOPE, INTERCEPT, CORREL and RSQ.
- 3 Predict the margin for a team that runs 30 times, and compute the residual for LAC.

The file is in weeks/week05/data/ after /update. Rush attempts are x and final margin is y. This is the same kind of line as the baseball one, on a season the Case Study does not use.



THE SAME TEN GAMES: STANDARD ERROR AND P

- 1 Open the second sheet, standard error and p.
- 2 Compute each residual, the residual standard error (check it with STEYX), and the standard error of the slope from DEVSQ.
- 3 Compute t, the p-value with T.DIST.2T and the 95% interval with T.INV.2T, then read the standard error out of LINEST.

The sheet ends with a judgment cell: would you tell a coach these ten games show rushing moves the margin? Say which number you leaned on.



RUNS PER GAME ON OBP AND SLG, 30 MLB TEAMS, 2024

- 1 Open the third sheet, two predictors.
- 2 Fit both predictors at once with LINEST and read the coefficients, standard errors and R squared out with INDEX.
- 3 Take each predictor's standard deviation with STDEV.S, multiply it by the coefficient, and say which of the two matters more.

LINEST(y range, both x columns, TRUE, TRUE). INDEX(..., 1, 2) is the OBP coefficient and INDEX(..., 1, 1) is the SLG coefficient: LINEST lists the x columns in reverse order.



CASE STUDY STEPS

- 1** Regress final margin on rush attempts across 2,174 team-games, 2022 to 2025, and read the output table.
- 2** Add the score the team was playing from, `mean_score_diff`, as a second predictor and compare the two tables.
- 3** Compute run share in each of five score states from the second data file, pooled and by season, and refit both regressions season by season.
- 4** Build an interactive scatter of the team-games with hover detail and a score-state or season control.

The data is `team-games-2022-2025.csv` and `plays-by-score-state-2022-2025.csv`. The README in `weeks/week05/` has the full list.



COACH MCDERMOTT WANTS TO KNOW "DOES RUNNING THE BALL WIN GAMES, OR DOES WINNING MAKE YOU RUN?"

- 1 Your first regression puts a number on what one more rush attempt is worth in final margin. Before you looked at anything else, how much of that number would you have believed, and why?
- 2 Once the score a team was playing from enters the model, what happens to the rushing coefficient, and what does that tell you about what the first regression was measuring?
- 3 Looking at run share by score state, which way does the arrow run between running the ball and winning? Say how the table supports your answer.
- 4 Something is left of the rushing coefficient after the control, and it still clears 0.05. Give two different explanations that would both produce that, and say what data would tell them apart.
- 5 An assistant on staff says the analytics say to run more. What would you tell him, in a sentence or two he would accept?
- 6 Which of the numbers you produced this week would you put in front of Coach McDermott, and which would you keep on your own screen? Why?

Your brief is written to Coach McDermott. `python3 scripts/new_brief.py week05` writes BRIEF.md with these as headings; `/coach-brief 5` critiques a draft and will not write one.

BEFORE YOU PUSH ON MONDAY, SEPTEMBER 28

Commit the filled-in worksheet in `weeks/week05/data/`.

Commit your analysis code and its outputs in `weeks/week05/outputs/`.

Answer every question in `BRIEF.md`, in your own words.

Run `/audit`, then `/quiz-me`.

Upload your repository's link to Canvas by Monday, September 28 at 11:59pm.